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## Unit – 4 : Session – 10 : SLO – 2 | Pipelining Performance — Fill in the Blanks

### ■ I. Fill in the Blanks

1. throughput — the primary pipelining metric, measured in instructions per cycle (IPC).
2. Pipeline stall — the pipeline pauses when dependencies cannot be resolved in time.
3. depth (number of stages) — ideal speedup equals the number of pipeline stages under perfect conditions.
4. Structural — hazards where multiple instructions need the same hardware resource.
5. Pipeline depth (ideal speedup) —  $\text{Speedup} = \text{Pipeline depth} / (1 + \text{pipeline stall cycles per instruction})$ .
6. Operand (result) — forwarding/bypassing reduces data hazards without waiting for register write-back.
7. Execute (EX) — branch conditions and target addresses are typically resolved in the execute stage.
8. Deep pipelines improve clock rate (throughput) but increase hazard/branch (stall) penalties.
9. Instruction-level — ILP lets multiple instructions execute in parallel within a single pipeline.
10. CPI — cycles per instruction, the average clock cycles each instruction needs.